P8109 Statistical Inference – Homework 5

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2026-04-30

Problem 1

A simple random sample of size $n = 30$ is drawn from a lot with unknown defective fraction p, where the prior on p is $\text{Uniform}(0, 1)$. Given that $X = 2$ defective items are observed, find the posterior probability $P(p \leq 0.1 \mid X = 2)$.

The likelihood from $X \mid p \sim \text{Binomial}(30, p)$ is

$$L(p \mid x) \;\propto\; p^x(1-p)^{n-x} \;=\; p^2(1-p)^{28}.$$

The $\text{Uniform}(0, 1)$ prior is $\pi(p) = 1 = \text{Beta}(1, 1)$. Then by "posterior $\propto$ prior $\times$ likelihood",

$$\pi(p \mid x) \;\propto\; p^{x+1-1}(1-p)^{n-x+1-1} \;=\; p^2(1-p)^{28},$$

which is the kernel of $\text{Beta}(\alpha^*, \beta^*)$ with $\alpha^* = x + 1 = 3$ and $\beta^* = n - x + 1 = 29$, i.e.,

$$p \mid X = 2 \;\sim\; \text{Beta}(3, 29).$$

Hence

$$P(p \leq 0.1 \mid X = 2) \;=\; F_{\text{Beta}(3,29)}(0.1).$$

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pbeta(0.1, shape1 = 3, shape2 = 29)
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## [1] 0.6114144
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So $P(p \leq 0.1 \mid X = 2) \approx 0.6114$.

Problem 2

Let $X \mid \theta \sim N(\theta, 100)$ (variance $\sigma_0^2 = 100$) and let the prior be $\theta \sim N(100, 225)$ (so $\xi = 100$, $\tau^2 = 225$). The observed score is $x = 115$. Show that (a) the posterior is $N(110.38, 69.23)$, and (b) the Bayes factor for $H_0 : \theta \leq 100$ versus $H_1 : \theta > 100$ is 0.12.

(a) Posterior distribution

With $n = 1$, the likelihood and prior are

$$f(x \mid \theta) \;=\; \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left\{-\frac{(x-\theta)^2}{2\sigma_0^2}\right\}, \qquad \pi(\theta) \;=\; \frac{1}{\sqrt{2\pi\tau^2}} \exp\left\{-\frac{(\theta-\xi)^2}{2\tau^2}\right\}.$$

By Example 10.1 (Lecture 10, slides 4–5), the posterior is $\theta \mid X \sim N(A, B^2)$ with

$$A \;=\; \frac{\dfrac{n\bar{X}}{\sigma_0^2} + \dfrac{\xi}{\tau^2}}{\dfrac{n}{\sigma_0^2} + \dfrac{1}{\tau^2}}, \qquad B^2 \;=\; \frac{1}{\dfrac{n}{\sigma_0^2} + \dfrac{1}{\tau^2}}.$$

Substituting $n = 1$, $x = 115$, $\sigma_0^2 = 100$, $\xi = 100$, $\tau^2 = 225$:

$$B^2 \;=\; \left(\frac{1}{100} + \frac{1}{225}\right)^{-1} \;=\; \frac{900}{13} \;\approx\; 69.23, \qquad A \;=\; B^2\left(\frac{115}{100} + \frac{100}{225}\right) \;=\; \frac{1435}{13} \;\approx\; 110.38.$$

Therefore

$$\theta \mid X = 115 \;\sim\; N(110.38,\; 69.23).$$

(b) Bayes factor

Both hypotheses are composite, so the simple-vs-simple BF formula does not apply. Using the General Composite Hypotheses formulation (Lecture 10, slide 15), the Bayes factor equals the ratio of posterior odds to prior odds:

$$\mathrm{BF} \;=\; \frac{p_0/p_1}{\pi_0/\pi_1}, \qquad p_0 \;=\; P(\theta \le 100 \mid X), \quad \pi_0 \;=\; P(\theta \le 100).$$

The prior is $N(100, 225)$, centered at 100, so $\pi_0 = 0.5$ and $\pi_0/\pi_1 = 1$. The posterior is $N(110.38, 69.23)$, so

$$p_0 \;=\; \Phi\!\left(\frac{100 - 110.38}{\sqrt{69.23}}\right) \;=\; \Phi(-1.247) \;\approx\; 0.1060,$$

and therefore

$$\mathrm{BF} \;=\; \frac{p_0/p_1}{\pi_0/\pi_1} \;=\; \frac{0.1060/0.8940}{1} \;\approx\; 0.12.$$

Problem 3

With $n = 20$ rabbits and $X \sim \mathrm{Binomial}(20, p)$, $X = 15$ survived. (a) Compute the posterior Bayes estimator of p under a $\mathrm{Beta}(3, 5)$ prior. (b) Obtain the MLE. (c) Recompute (a) with a $\mathrm{Uniform}(0, 1)$ prior. (d) For $n = 200$, $x = 150$, compare the MLE, the $\mathrm{Beta}(3, 5)$ posterior mean, and the Uniform-prior posterior mean.

For Beta–Binomial conjugacy (Lecture 10, Example 10.4 / Example 10.9, slide 38),

$$L(p \mid x) \;\propto\; p^x(1-p)^{n-x}, \qquad \pi(p) \;\propto\; p^{\alpha-1}(1-p)^{\beta-1},$$
$$\pi(p \mid x) \;\propto\; p^{\alpha+x-1}(1-p)^{\beta+n-x-1}, \qquad p \mid X \;\sim\; \mathrm{Beta}(\alpha + x,\; \beta + n - x).$$

Under squared error loss the Bayes estimator is the posterior mean (Corollary 10.1(i), slide 29):

$$\hat{p}_{\mathrm{Bayes}} \;=\; \mathrm{E}[p \mid X] \;=\; \frac{\alpha + x}{\alpha + \beta + n}.$$

(a) Beta(3,5) prior, $n = 20$, $x = 15$

$$p \mid X \;\sim\; \mathrm{Beta}(3 + 15,\; 5 + 5) \;=\; \mathrm{Beta}(18,\; 10),$$
$$\hat{p}_{\mathrm{Bayes}} \;=\; \frac{18}{28} \;=\; \frac{9}{14} \;\approx\; 0.64.$$

(b) MLE

$$\hat{p}_{\text{MLE}} \;=\; \frac{X}{n} \;=\; \frac{15}{20} \;=\; 0.75.$$

(c) Uniform = Beta(1,1) prior

$$p \mid X \;\sim\; \text{Beta}(1+15,\; 1+5) \;=\; \text{Beta}(16,\, 6),$$
$$\hat{p}_{\text{Bayes}} \;=\; \frac{16}{22} \;=\; \frac{8}{11} \;\approx\; 0.73.$$

The estimate increases from 0.64 to 0.73 when the more informative prior $\text{Beta}(3,5)$ (which puts mass on smaller p) is replaced by the flat prior.

(d) $n = 200$, $x = 150$

$$\hat{p}_{\text{MLE}} \;=\; \frac{150}{200} \;=\; 0.75, \qquad \hat{p}_{\text{Beta}(3,5)} \;=\; \frac{153}{208} \;\approx\; 0.74, \qquad \hat{p}_{\text{Unif}} \;=\; \frac{151}{202} \;\approx\; 0.75.$$

Comment. All three estimators are essentially equal at $n = 200$. As n grows, the data terms dominate the fixed prior parameters in $(\alpha + x)/(\alpha + \beta + n)$, so the posterior mean converges to the MLE x/n regardless of the prior. The discrepancies seen at $n = 20$ between 0.64 (Beta(3,5)) and 0.75 (MLE) shrink to less than one percentage point at $n = 200$.

Problem 4

Let $X \mid \theta \sim \text{Gamma}(\alpha, 1/\theta)$ with density $f(x \mid \theta) = \frac{\theta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\theta x}$ and let $\theta \sim \text{Gamma}(\beta, 1/\gamma)$ with density $\pi(\theta) = \frac{\gamma^\beta}{\Gamma(\beta)} \theta^{\beta-1} e^{-\gamma\theta}$. Obtain the unconditional density of X.

By the marginal density formula (Lecture 10, slide 3),

$$f(x) \;=\; \int_0^\infty f(x \mid \theta)\, \pi(\theta)\, d\theta \;=\; \int_0^\infty \frac{\theta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\theta x} \;\cdot\; \frac{\gamma^\beta}{\Gamma(\beta)} \theta^{\beta-1} e^{-\gamma\theta}\, d\theta.$$

Pulling constants outside,

$$f(x) \;=\; \frac{x^{\alpha-1}\, \gamma^\beta}{\Gamma(\alpha)\Gamma(\beta)} \int_0^\infty \theta^{\alpha+\beta-1}\, e^{-(x+\gamma)\theta}\, d\theta.$$

The integrand is the kernel of a $\text{Gamma}(\alpha + \beta,\, \text{rate} = x + \gamma)$ density in θ, so by

$$\int_0^\infty \theta^{a-1} e^{-b\theta}\, d\theta \;=\; \frac{\Gamma(a)}{b^a},$$

we obtain

$$\int_0^\infty \theta^{\alpha+\beta-1}\, e^{-(x+\gamma)\theta}\, d\theta \;=\; \frac{\Gamma(\alpha+\beta)}{(x+\gamma)^{\alpha+\beta}}.$$

Therefore

$$f(x) \;=\; \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\, \frac{\gamma^\beta\, x^{\alpha-1}}{(x+\gamma)^{\alpha+\beta}}, \qquad x > 0.$$

This is a (scaled) Beta-prime / compound-Gamma form.

Problem 5

Let θ have the Pareto-type prior $\pi(\theta) = \gamma\beta^\gamma\theta^{-(\gamma+1)}$ for $\theta \geq \beta$, with $\beta > 0$ and $\gamma > 2$. (a) Find the prior mean of θ. (b) Given $Y_1, \dots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta)$, find the posterior mean of θ. (c) Show that $\widehat{\theta} = \frac{\gamma+n}{\gamma+n-1} \max\{z, \beta\}$, with $z = \max(y_1, \dots, y_n)$, is admissible under squared error loss.

(a) Prior mean

$$\mathrm{E}[\theta] \;=\; \int_\beta^\infty \theta \cdot \gamma\beta^\gamma\theta^{-(\gamma+1)}\,d\theta \;=\; \gamma\beta^\gamma \int_\beta^\infty \theta^{-\gamma}\,d\theta.$$

Provided $\gamma > 1$ (which holds since $\gamma > 2$),

$$\int_\beta^\infty \theta^{-\gamma}\,d\theta \;=\; \left[\frac{\theta^{-\gamma+1}}{-\gamma+1}\right]_\beta^\infty \;=\; \frac{\beta^{1-\gamma}}{\gamma-1}.$$

Hence

$$\mathrm{E}[\theta] \;=\; \gamma\beta^\gamma \cdot \frac{\beta^{1-\gamma}}{\gamma-1} \;=\; \frac{\gamma\beta}{\gamma-1}.$$

(The first moment requires $\gamma > 1$; the assumption $\gamma > 2$ guarantees the second moment, hence the variance, is also finite.)

(b) Posterior mean

For $Y_i \mid \theta \sim \text{Uniform}(0, \theta)$, each density is $\theta^{-1}\mathbf{1}\{0 < y_i < \theta\}$, so with $z = \max(y_1, \dots, y_n)$,

$$L(\theta \mid \mathbf{y}) \;=\; \theta^{-n}\,\mathbf{1}\{\theta \geq z\}.$$

Combining with the prior, which is supported on $\theta \geq \beta$,

$$\pi(\theta \mid \mathbf{y}) \;\propto\; \theta^{-n}\mathbf{1}\{\theta \geq z\} \cdot \theta^{-(\gamma+1)}\mathbf{1}\{\theta \geq \beta\} \;=\; \theta^{-(n+\gamma+1)}\mathbf{1}\{\theta \geq \max(z, \beta)\}.$$

Let $m = \max(z, \beta)$. The normalizing constant is

$$\int_m^\infty \theta^{-(n+\gamma+1)}\,d\theta \;=\; \frac{m^{-(n+\gamma)}}{n+\gamma},$$

so

$$\pi(\theta \mid \mathbf{y}) \;=\; (n+\gamma)\,m^{n+\gamma}\,\theta^{-(n+\gamma+1)}, \qquad \theta \geq m,$$

which is a Pareto kernel of the same family with shape $n + \gamma$ and lower endpoint m. The posterior mean (provided $n + \gamma > 1$) is

$$\mathrm{E}[\theta \mid \mathbf{y}] \;=\; \int_m^\infty \theta \cdot (n+\gamma)\,m^{n+\gamma}\theta^{-(n+\gamma+1)}\,d\theta \;=\; (n+\gamma)m^{n+\gamma} \int_m^\infty \theta^{-(n+\gamma)}\,d\theta.$$

Evaluating the integral,

$$\int_m^\infty \theta^{-(n+\gamma)}\,d\theta \;=\; \frac{m^{1-(n+\gamma)}}{n+\gamma-1},$$

so

$$\mathrm{E}[\theta \mid \mathbf{Y}] \;=\; \frac{n+\gamma}{n+\gamma-1}\,\max(z, \beta).$$

(c) Admissibility

The given estimator is exactly the posterior mean from part (b):

$$\hat{\theta} \;=\; \frac{\gamma+n}{\gamma+n-1}\,\max\{z,\beta\} \;=\; \mathrm{E}[\theta \mid \mathbf{Y}].$$

By Corollary 10.1(i) (Lecture 10, slide 29), under squared error loss the Bayes estimator is the posterior mean. Hence $\hat{\theta}$ is the Bayes estimator with respect to the prior $\pi(\theta) = \gamma\beta^{\gamma}\theta^{-(\gamma+1)}$.

The posterior is a proper, non-degenerate Pareto distribution, so the posterior mean – and therefore the Bayes estimator under squared error loss – is unique. By the lecture result that *a Bayes estimator, when unique, is always admissible* (Lecture 10, slide 26), we conclude that $\hat{\theta}$ is admissible under squared error loss.

Problem 6

Crash probabilities are 5×10^{-6} for plane and 10^{-2} for car. The loss table is plane: 10^{6} if crash, 5 if no crash; car: 100 if crash, 0 if no crash. Which decision has smaller (Bayes) risk?

Each decision induces its own prior over $\theta \in \{\text{crash}, \text{no crash}\}$. The Bayes risk is the expected loss under the mode-specific crash probability:

$$r(d_{\text{plane}}) \;=\; 10^{6}\cdot(5\times10^{-6}) + 5\cdot(1-5\times10^{-6}) \;\approx\; 10,$$

$$r(d_{\text{car}}) \;=\; 100\cdot(10^{-2}) + 0\cdot(1-10^{-2}) \;=\; 1.$$

Conclusion. The car has the smaller Bayes risk (1 vs. ≈ 10). Despite the much higher per-crash loss for an airplane crash, the much lower crash probability for the plane is more than offset by the everyday loss of 5 assigned to a non-crash plane trip versus 0 for a non-crash car trip. Under this loss specification, the car is the Bayes decision.

Problem 7

Risk table on $\Theta = \{\theta_1,\theta_2,\theta_3,\theta_4\}$ with prior $\pi = (1/8, 3/8, 1/4, 1/4)$ and decisions $\{d_1, d_2, d_3\}$:

	d_1	d_2	d_3
θ_1	0	2	3
θ_2	1	0	2
θ_3	3	4	0
θ_4	1	2	0

(a) Bayes risk of each decision; (b) which decision is Bayes; (c) the minimax decision(s).

(a) Bayes risk

Using $r(d_j) \;=\; \sum_{i=1}^{4} R(d_j,\theta_i)\,\pi(\theta_i)$,

$$r(d_1) \;=\; 0\cdot\tfrac{1}{8} + 1\cdot\tfrac{3}{8} + 3\cdot\tfrac{1}{4} + 1\cdot\tfrac{1}{4} \;=\; \tfrac{11}{8} \;=\; 1.375,$$

$$r(d_2) \;=\; 2\cdot\tfrac{1}{8} + 0\cdot\tfrac{3}{8} + 4\cdot\tfrac{1}{4} + 2\cdot\tfrac{1}{4} \;=\; \tfrac{7}{4} \;=\; 1.75,$$

$$r(d_3) \;=\; 3\cdot\tfrac{1}{8} + 2\cdot\tfrac{3}{8} + 0\cdot\tfrac{1}{4} + 0\cdot\tfrac{1}{4} \;=\; \tfrac{9}{8} \;=\; 1.125.$$

(b) Bayes decision

The minimum Bayes risk is $r(d_3) = 9/8 = 1.125$, so Therefore d_3 is the Bayes decision.

(c) Minimax decision

The maximum risk for each decision (column maxima of the risk table) is

$$\max_i R(d_1, \theta_i) = 3, \qquad \max_i R(d_2, \theta_i) = 4, \qquad \max_i R(d_3, \theta_i) = 3.$$

The minimum of these column maxima is 3, attained by both d_1 and d_3. Therefore the **minimax decisions are d_1 and d_3 (tie).**

Problem 8

Let $X \sim \text{Bernoulli}(\theta)$ and consider an estimator $d(X)$ with $d(1) = d_1$, $d(0) = d_2$, under squared error loss $L(d, \theta) = (d - \theta)^2$. (a) Write the risk function $R_d(\theta)$. (b) Show that the minimax decision satisfies $d_1 = 3/4$, $d_2 = 1/4$.

(a) Risk function

$$R_d(\theta) \;=\; \mathop{\mathrm{E}}_{\theta}[(d(X) - \theta)^2] \;=\; (d_1 - \theta)^2\, P(X = 1 \mid \theta) + (d_2 - \theta)^2\, P(X = 0 \mid \theta),$$

$$R_d(\theta) \;=\; \theta(d_1 - \theta)^2 + (1 - \theta)(d_2 - \theta)^2.$$

(b) Minimax decision via Theorem 10.2

We follow the constant-risk Bayes route (Theorem 10.2, Lecture 10, slide 33), mirroring Example 10.9 for $n = 1$.

Step 1: Bayes estimator under a Beta(α, β) prior. With $X \sim \text{Bernoulli}(\theta)$ and prior $\theta \sim \text{Beta}(\alpha, \beta)$, the conjugate posterior is $\theta \mid X \sim \text{Beta}(\alpha + X, \beta + 1 - X)$. The Bayes estimator under squared error loss is the posterior mean:

$$T(X) \;=\; \frac{X + \alpha}{\alpha + \beta + 1}.$$

Step 2: Risk as a polynomial in θ. Substituting $d_1 = T(1) = \dfrac{1 + \alpha}{\alpha + \beta + 1}$ and $d_2 = T(0) = \dfrac{\alpha}{\alpha + \beta + 1}$ into $R_d(\theta)$ and writing $c = \alpha + \beta + 1$,

$$R_T(\theta) \;=\; \mathop{\mathrm{Var}}_{\theta}(T(X)) + (\mathop{\mathrm{E}}_{\theta}[T(X)] - \theta)^2.$$

With $\mathrm{E}_\theta[T(X)] = \dfrac{\theta + \alpha}{c}$ and $\mathrm{Var}_\theta(T(X)) = \dfrac{\theta(1 - \theta)}{c^2}$,

$$R_T(\theta) \;=\; \frac{\theta(1 - \theta)}{c^2} + \left(\frac{\theta + \alpha}{c} - \theta\right)^2 \;=\; \frac{\theta(1 - \theta)}{c^2} + \frac{(\alpha - (c - 1)\theta)^2}{c^2}.$$

Multiplying through by c^2 and collecting powers of θ,

$$c^2 R_T(\theta) \;=\; \alpha^2 \;+\; [1 - 2\alpha(c - 1)]\theta \;+\; [(c - 1)^2 - 1]\theta^2.$$

Step 3: Solve for constant risk. For $R_T(\theta)$ to be constant in θ, set the coefficients of θ and θ^2 to zero:

$$(c-1)^2 - 1 \;=\; 0 \implies \alpha + \beta \;=\; 1, \qquad 1 - 2\alpha \;=\; 0 \implies \alpha \;=\; \beta \;=\; \tfrac{1}{2}.$$

With $\alpha = \beta = 1/2$ and $c = 2$,

$$d_1 \;=\; T(1) \;=\; \frac{1 + 1/2}{2} \;=\; \frac{3}{4}, \qquad d_2 \;=\; T(0) \;=\; \frac{1/2}{2} \;=\; \frac{1}{4}.$$

Step 4: Conclude. The Bayes estimator $T^*(X) = (X + 1/2)/2$, corresponding to $(d_1, d_2) = (3/4, 1/4)$, has constant risk

$$R_{T^*}(\theta) \;=\; \frac{\alpha^2}{c^2} \;=\; \frac{1/4}{4} \;=\; \frac{1}{16}.$$

By Theorem 10.2, a Bayes estimator with constant risk is minimax. Therefore the minimax decision satisfies

$$d_1 \;=\; \tfrac{3}{4}, \qquad d_2 \;=\; \tfrac{1}{4}.$$